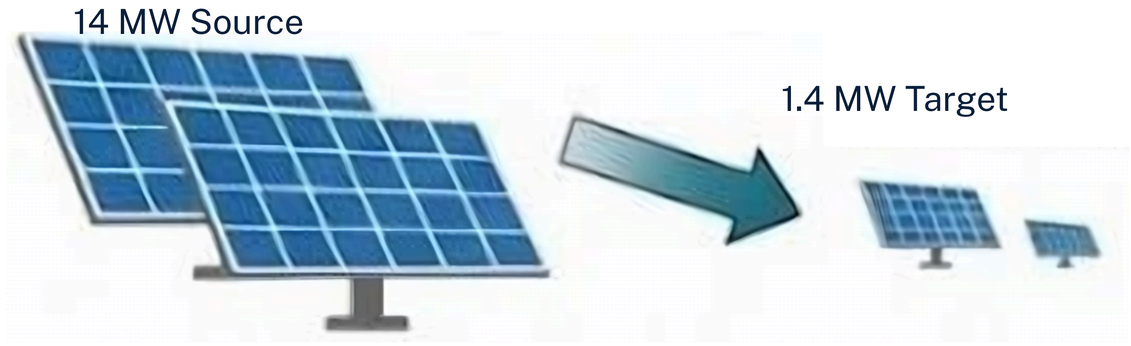
LEARNING WHEN TO TRANSFER: ROBUST SOLAR POWER FORECASTING UNDER CAPACITY MISMATCH

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1. PROBLEM STATEMENT & MOTIVATION

Cold-start PV data scarcity and magnitude overfitting during transfer between plants of unequal capacity



Magnitude overfitting under unequal capacities

Objectives

- Investigate zero-target-label unsupervised transfer
- Decouple scale via per-unit normalization
- Eliminate deterministic diurnal cycles with the Clear-Sky Index
- Align dynamics through a CDAN

2. KEY CONTRIBUTIONS & DATA

1) Failure Mode Characterization

- Identifies magnitude overfitting as the cause of transfer collapse and motivates magnitude normalization

2) Physics-Guided Domain Alignment

- Per-unit scaling and the Clear-Sky Index remove capacity and diurnal effects before CDAN alignment

3) Transfer Boundary Delineation

(mature source) (cross-continental target with discordant hemispheric seasonality)

- Tests intra-regional and cross-continental transfer

3. PROPOSED METHODOLOGY & ARCHITECTURE

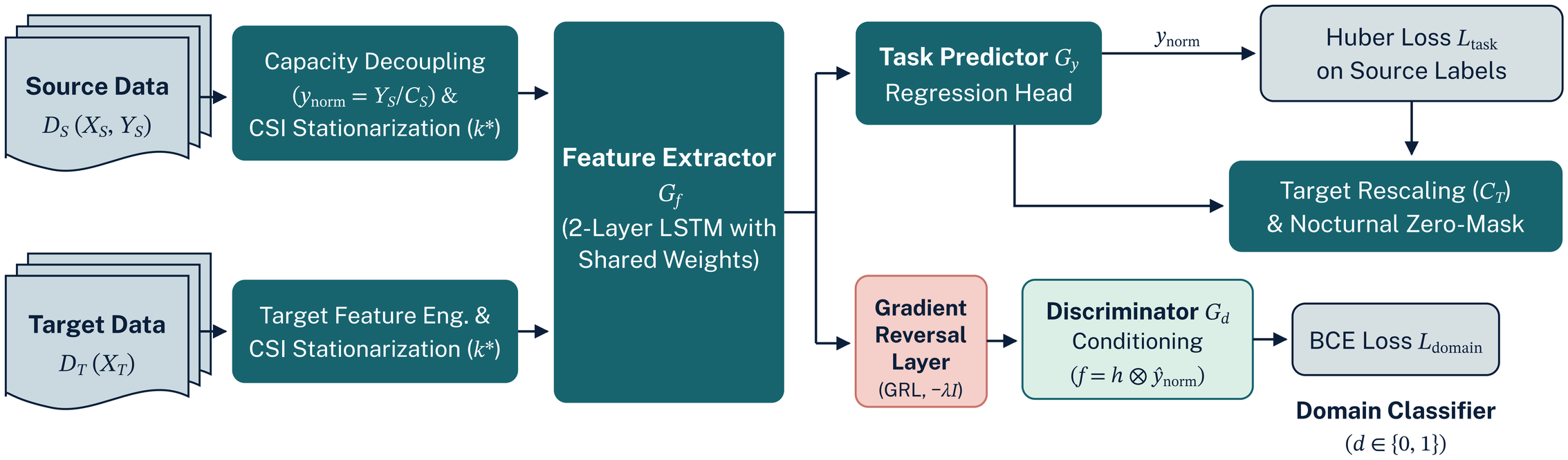


Fig. 1. Proposed CDAN + LSTM + CSI architecture

4. EXPERIMENTAL BENCHMARK & TRACKING

	Task 1 India	Task 2 Australia	Task 3 Australia
A1: Source-only LSTM	0.8325	–	–
A3: CDAN+LSTM	0.8329	–	–
A4: Transformer	–	0.5428	0.5428
Proposed (CDAN+LSTM+CSI)	**0.8325**	**0.5428**	**–**

Uncalibrated Baseline: $\hat{Y}$ > 10,000 kW (Off-Scale)

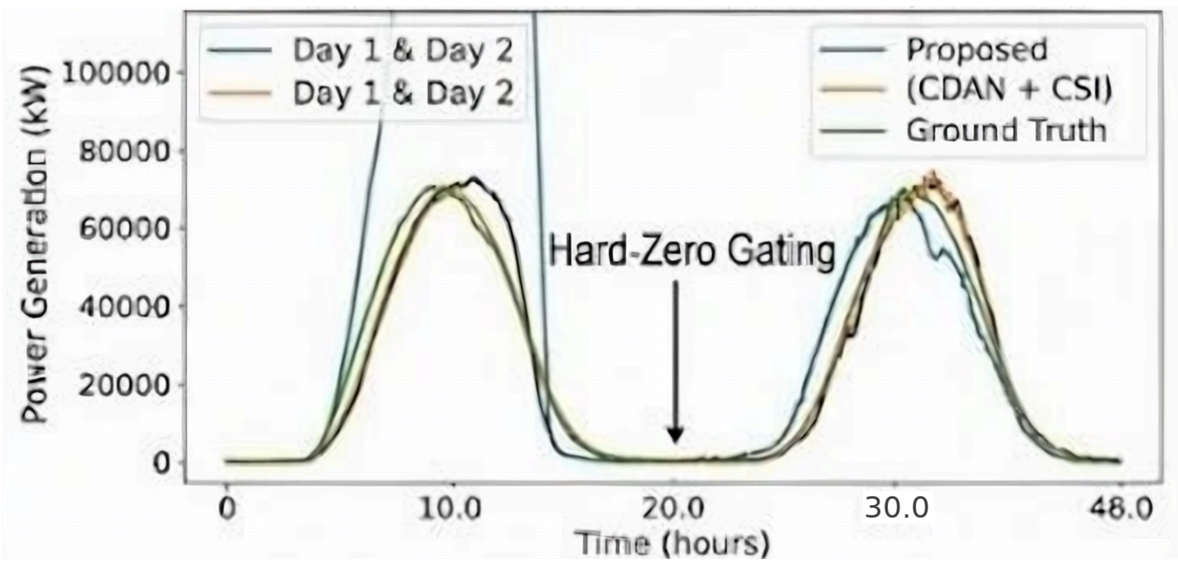


Fig. 2. 48-hour tracking curve

5. DISCUSSION & FUTURE WORK

Key Insight

- **Adversarial alignment** is highly effective **intra-regionally**, but **cross-continental transfer** (India to Australia) encounters **negative transfer ($R^2 < 0$)** due to **seasonal phase misalignment**.
- Unadapted Transformers are resilient and retain positive skill.



Intra-Regional CDAN (Aligned Representation)

Cross-Hemisphere CDAN (Negative Transfer)

Conclusion

- Resolves the uncalibrated magnitude collapse with per-unit scaling and nocturnal zero-masking.
- **Outlines** adaptation boundaries: intra-regional alignment helps, cross-continental transfer does not.
- **Establishes a zero-label transfer benchmark.**
- **Future work with post-commissioning fine-tuning.**
- **Acknowledgment:** GenAI/LLM tools helped in preparing this poster.